SMART INDIA HACKATHON 2026 • OFFICIAL PRESENTATION DOSSIER

COOPSERVE

The Cooperative Service Bridge

A Worker-Owned Cooperative Gig Platform connecting urban households with verified local tradespeople—transforming exploited gig laborers into dignified member co-owners.

Problem Statement ID: SIH26089	**Team Name:** PIXEL PIONEERS
Theme: Smart Automation	**Category:** Software
Title: Cooperative Gig Services Platform for Household & Community Services	**Live Demo:** http://localhost:3000
Git Repository: github.com/jeshwanth-08/coopserve1	**Active Branch:** coopserve3

The 60-Second Verbal Pitch (Memorize and deliver with passion):
"Honorable judges, private gig aggregators like Urban Company and TaskRabbit treat skilled workers as replaceable commodities—taking 25% to 30% commissions, using opaque black-box algorithms, and offering zero safety net. Meanwhile, India's registered Labour Cooperative Federations have millions of certified electricians, plumbers, and technicians who remain digitally invisible.

Introducing **CoopServe**—India's first worker-owned cooperative gig platform. On CoopServe, skilled tradespeople are not exploited gig workers—they are verified member co-owners who take home **85%+ of job earnings**, contribute to a pooled **welfare and health fund**, and receive work through **fair algorithmic rotation** rather than favoritism.

For households, we provide verified local trust, instant visual AI damage diagnostics, bundled service packages, and transparent pricing. CoopServe bridges the digital divide for India's cooperative sector."

1. Key Differentiators: CoopServe vs. Private Platforms

When the jury compares CoopServe to existing players like Urban Company, emphasize these fundamental structural differences:

Dimension	Urban Company / Private Aggregators	CoopServe (Cooperative Model)
Worker Relationship	Independent contractors; zero ownership or voting voice	**Member co-owners** with democratic voting rights & shared surplus
Platform Commission	Aggressive **25% to 30%** commission extracted to private VCs	**8% to 10%** cooperative operational cost recovery
Worker Take-Home	Diminished payout (~65% after penalties & fees)	**85%+ direct take-home pay** directly to the worker
Social Security	None; workers bear all personal accident and health risk	**5% Cooperative Welfare Pool** + transit damage insurance
Work Allocation	Opaque algorithm maximizing corporate margins	**Fair Work Rotation** (prevents monopolization by cliques)
Pricing & Surges	Dynamic surge pricing + hidden consumable markups	Standardized cooperative rates; **zero surge pricing**
Dispute Resolution	Unilateral automated penalties against workers	**Cooperative Node Peer Committee** with escrow review

2. End-to-End Solution Lifecycle

1. Register & Request: Households request services via search, category browsing, or 1-click curated package bundles.	**2. AI Visual Problem Detection:** User uploads a photo of damage. Gemini Vision analyzes the image, root cause, and severity.
3. Choose & Schedule: Leaflet + OSM canvas selects verified specialists. Instant choice of slots or 45-min ASAP emergency.	**4. Real-Time Parallel Execution:** Provider triggers "On The Way" and "In Progress"—customer screen updates in parallel.
5. Multi-Mode Payment: Zero-fee checkout via UPI (GPay/PhonePe/QR), Cards, NetBanking, or Cash After Service with GST invoice.	**6. Rate, Review & Escrow:** Customer rates service. Escrow automatically splits: 85% worker, 5% welfare pool, 10% operations.

3. Step-by-Step Live Demo Script (5 Minutes)

Run this exact chronological demo sequence on **http://localhost:3000** to captivate the judges:

Step 1: Homepage & Cooperative Trust (1 Min)

• Open **http://localhost:3000**.
• Point to the header: "CoopServe — The Cooperative Service Bridge". Explain the worker co-ownership model.
• Demonstrate the **Quick Booking Modal** with multi-service option selection.
• Show that unauthenticated bookings automatically redirect to sign in with safe return URLs.

Step 2: AI Visual Problem Detection — The Showstopper (1.5 Mins)

• Navigate to **/ai-diagnosis**.
• Upload an image of an issue (e.g. AC frost buildup, pipe leakage, socket burnout).
• Show the judges: Multimodal AI analyzes the image, identifies the root cause, provides estimated repair costs (■), safety alerts, and pre-fills the booking form.
• *Key Highlight:* Non-maintenance pictures (quotes, random text) are automatically rejected by vision filters with friendly guidance!

Step 3: Curated Packages & Interactive Leaflet Map (1 Min)

• Navigate to **/packages**.
• Click "Book Package" on the **Home Refresh Package** (■1999) or **Summer AC Double Shield** (■999).
• Show that it loads the exact package details, bundle price, and included services checklist (not defaulting to an electrician!).
• Open the **Interactive Leaflet Map canvas** with OpenStreetMap tiles and GPS pin selection.
• Point out the ■ **45-Minute Emergency ASAP** booking toggle.

Step 4: Real-Time Parallel Status Sync (1 Min)

• Open two browser windows side-by-side: Member Dashboard (**/member**) and Provider Jobs (**/provider**).
• In the provider window, click **"On The Way"**.
• *Watch the customer screen update simultaneously in parallel without any page reload!*
• Point out the glowing green **Live Specialist Beacon** with estimated arrival and 1-click call specialist button.

Step 5: Cooperative Control Center / Admin HQ (30 Secs)

• Open **/admin**.
• Show the 7 Core KPI cards (Total Customers, Active Pros, Completed Jobs, GMV, Average Rating, Resolution Rate, Society Pools).
• Show the **Demand by Discipline** chart, Bookings Matrix, and Dispute Moderation tabs.

4. Technical Architecture & Tech Stack

Layer	Technologies Used	Core Responsibility
Frontend PWA	Next.js 14 App Router, React 18, Tailwind CSS, Lucide Icons	Mobile-first responsive UI, MobileBottomNav, booking stepper, real-time status banners
Mapping & Geo-Spatial	Leaflet.js + OpenStreetMap (OSM)	Interactive draggable map canvas, locality coordinates mapping, society cluster radius
AI & Intelligence	Google Gemini 1.5 Flash Multimodal Vision + TF-IDF	Visual damage detection, severity classification, parts estimation, search intent
Data & Backend	Next.js API Routes, Prisma ORM, SQLite / PostgreSQL	CRUD operations, JWT role-based access control, schema migrations, seed data
Real-Time Sync	BroadcastChannel API + Background Polling	Cross-tab and cross-device zero-latency parallel status synchronization
Payments & Invoicing	Simulated 256-bit Modal + Razorpay Webhooks	UPI, Cards, NetBanking, Wallets, Cash After Service, GST calculation, PDF receipt
Deployment	Cloudflare Global Edge CDN Tunnel	TLS/SSL termination, sub-second latency from Mumbai (BOM) edge nodes, global access

5. Cooperative Economics: The Math That Wins SIH

Judges love tangible financial comparisons. Here is the exact economic comparison for a ■1,000 household service:

Extractive Platform (e.g. Urban Company)

- Customer Pays: **■1,000**
- Platform Commission (28%): **-■280**
- Mandatory Consumable Markup: **-■70**
- Worker Health/Insurance: **■0 (None)**
- **Worker Net Take-Home: ~■650 (65%)**
- *Surplus Destination: Extracted to private VC funds*

CoopServe Cooperative Model

- Customer Pays: **■1,000**
- Direct Worker Take-Home (85%): **■850**
- Cooperative Welfare & Insurance Pool (5%): **■50**
- Platform Operational Maintenance (10%): **■100**
- **Year-End Patronage Dividend:** Shared surplus redistributed
- *Surplus Destination: Circulates in the local district economy*

Regulatory Backing: Built to comply with the **Multi-State Co-operative Societies (MSCS) Act, 2002** and directly aligns with recommendations from the **NITI Aayog 2022 Report on the Gig & Platform Economy**.

6. Multi-Stakeholder Impact Matrix

Stakeholder	Measurable Tangible Impact
Households	Background-verified cooperative professionals; standardized fair rates with zero surge pricing; visual AI damage estimates before booking; 30-day rework satisfaction guarantee.
Service Providers	Dignity as member co-owners rather than dispensable gig workers; 85%+ earnings share; emergency welfare fund; transparent fair work rotation; free trade upskilling.
Local Communities	Preserves local service capital inside municipal wards; strengthens neighbourhood trust networks via Apartment Society Pools; eliminates corporate intermediaries.
Environment	Hyper-local geo-matching minimizes worker commute distances from 15–20 km down to 2–5 km, cutting fuel consumption and urban carbon footprints per job.

7. Top 8 Tough Judge Questions & Winning Answers

Q1: "Urban Company already has massive market share. Why would customers choose CoopServe?"

Answer: "Consumers are increasingly frustrated with Urban Company's aggressive price surges, hidden consumable fees, and worker strikes. CoopServe delivers standardized cooperative rates with zero surge pricing, verified community accountability through housing society pools, and consumers feel proud knowing 85%+ of their money goes directly to the worker."

Q2: "How does your Fair Work Rotation work? Won't top-rated workers complain?"

Answer: "Unlike private platforms that funnel 80% of jobs to a top 10% clique, our algorithm uses a weighted rotation score: Score = f(Rating, Proximity, Time Since Last Job, Category Match). This ensures every verified provider gets steady work while maintaining high service standards. Furthermore, surplus profits are redistributed as dividends, rewarding high-performing members collectively."

Q3: "How does your AI Problem Detection actually work?"

Answer: "When a user uploads a photo, our system calls Google Gemini 1.5 Flash multimodal vision with specialized prompt engineering. The model extracts physical indicators (corrosion, burn marks, frost, moisture), estimates fault severity, specifies necessary replacement parts, and maps the issue to the exact cooperative discipline."

Q4: "How do you handle disputes if a customer claims shoddy work or a worker claims unfair treatment?"

Answer: "We feature a built-in Disputes & Complaints Matrix in our Federation Admin Dashboard. Payment is held in milestone escrow. When a dispute is raised, photo evidence before and after the job is reviewed by the local cooperative node manager rather than an automated algorithm that blindly penalizes workers."

Q5: "How does the cooperative federation adopt this technology if workers are non-technical?"

Answer: "The provider app is designed with a minimal touch-first interface featuring audio-visual prompts, WhatsApp SMS dispatch integration, and 1-tap action buttons ('On The Way', 'Start Work'). Furthermore, regional cooperative hubs provide in-person onboarding and digital literacy support."

Q6: "How do you scale this nationally?"

Answer: "We use a federated node model. Instead of building physical branches, we partner with registered state and district Labour Cooperative Federations. Each cooperative node manages local worker verification and disputes, while CoopServe provides the unified digital cloud platform."